MILITARY STANDARD

MIL-STD-1320-110
(NAVY)

10 JULY 1973

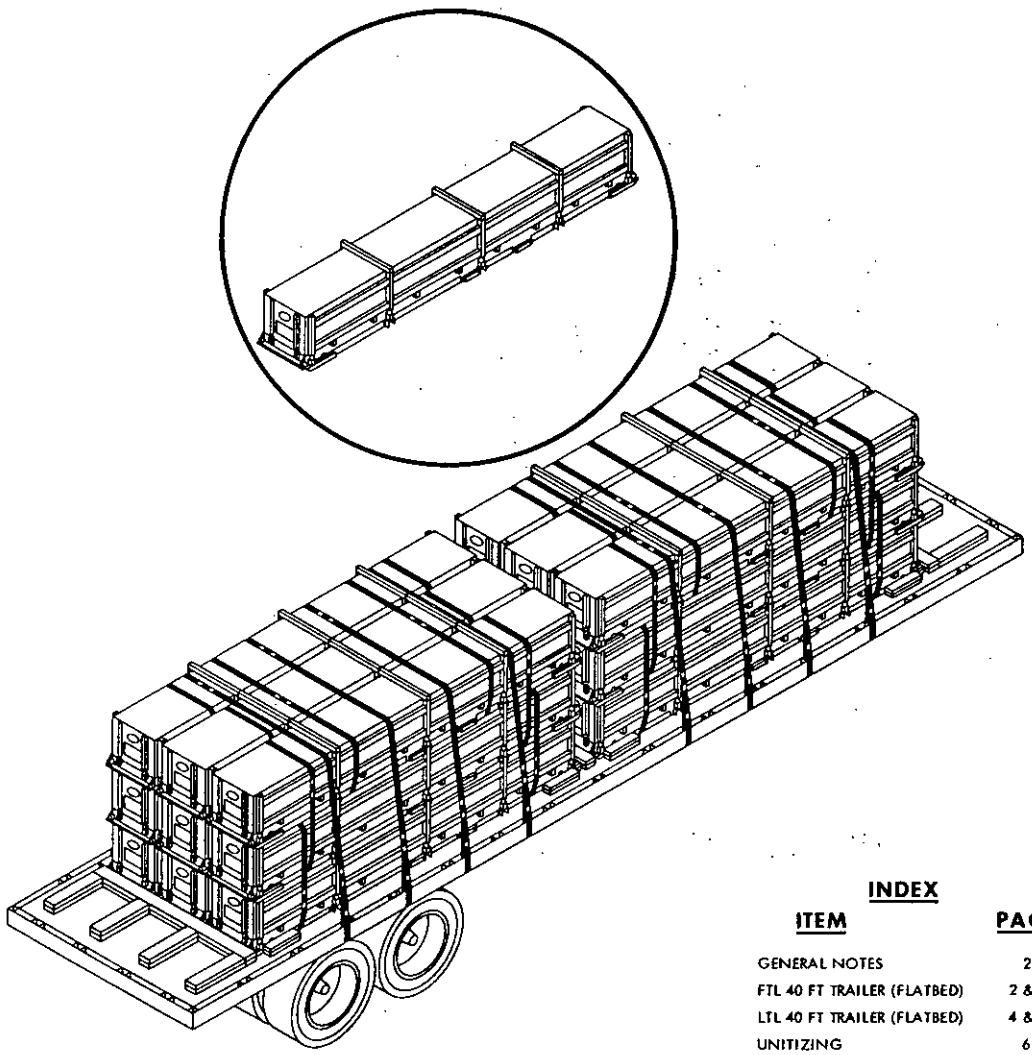
TRUCKLOADING

STANDARD ARM MISSILE AGM-78
IN CONTAINERS MK 372 MOD 4 & 6 OR CNU-183/E

CONTAINER DATA

SEE TABLE I

DOT HAZARD CLASSIFICATION . . . EXPLOSIVES A



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NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINER LOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M.S. Gray NWHL 6/18/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 6/18/73
SIGNATURE (AIR) SYSCOM BY DIRECTION DATE

ORIGINATOR Charles McBride 6/18/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE STANDARD ARM MISSILE AGM-78 IN THE MK 372 MOD 4 & 6 AND CNU-183/E CONTAINERS.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS. THE TRAILER AXLES SHALL BE LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
3. WHEN CONTAINERS ARE TWO OR THREE HIGH, THEY SHALL BE UNITIZED AS SHOWN ON PAGE 6.

WARNING

TO AVOID TOPPLING, SPECIAL CARE SHOULD BE TAKEN WHEN MOVING UNITIZED CONTAINERS. THE UNITIZING OF THE CONTAINERS IS AUTHORIZED AS PART OF THE TRUCKLOADING PROCEDURE ONLY. UNITIZING IS TO ENSURE CONTINUOUS ENGAGEMENT OF THE STACKING FEATURES AND MOVEMENT OF UNITIZED CONTAINERS SHOULD BE LIMITED TO THAT NECESSARY TO LOAD OR UNLOAD THE TRAILER.

4. LOAD BINDERS AND CHAINS MAY BE USED AS AN ALTERNATE FOR THE 2 INCH STRAPPING SUBSTITUTING ONE CHAIN AND LOAD BINDER FOR EACH 2 INCH STRAP. LAMINATED DOUBLED 2 x 6's SHALL BE USED OVER THE TOP OF THE CONTAINERS TO PREVENT DAMAGE BY THE CHAIN AND THE CHAIN SHALL BE SECURED TO THE BOARDS WITH FOUR 20d NAILS BY DRIVING THE NAIL THROUGH THE CHAIN LINK INTO THE BOARD AND CLINCHING THE NAIL OVER THE CHAIN.
5. CHAINS AND LOAD BINDERS SHALL MEET THE FOLLOWING CONDITIONS:
   A. CHAIN AND FITTINGS SHALL BE 3/8 INCH AND LOAD BINDERS 3/8 INCH CAPACITY.
   B. CHAIN, FITTINGS, AND LOAD BINDERS SHALL HAVE A MINIMUM SAFE WORKING LOAD OF 5000 LBS.
   C. LOAD BINDERS SHALL BE SAFETY WIRED WITH 16 GAUGE SOFT ANNEALED IRON WIRE OR SECURED USING THE END PIECE OF THE 3/8 CHAIN.
   PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY NAVWEPS OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS, AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.
6. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS, AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
7. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA AND ATTACH "EXPLOSIVES A" PLACARD TO BOTH SIDES, FRONT, AND BACK OF TRAILER.
8. APPLICABLE MATERIAL SPECIFICATIONS:
   DUNNAGE LUMBER, MM-L-751;
   NAILS FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT;
   STRAPPING QQ-S-781, TYPE I, CLASS A.
9. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL MILITARY STANDARD, MIL-STD-1320 (NAVY).

FTL 40 FT. TRAILER (FLATBED)

1. EACH STACK OF CONTAINERS SHALL BE SECURED TOGETHER WITH TWO 1 1/4 x .035 CROSS STRAPS. THE STRAPS PASS OVER THE COVERS OF THE CONTAINERS IN THE TOP LAYER AND RETURN PASSING UNDER THE BOTTOMS OF THESE CONTAINERS, BINDING THE TOP LAYER TOGETHER.

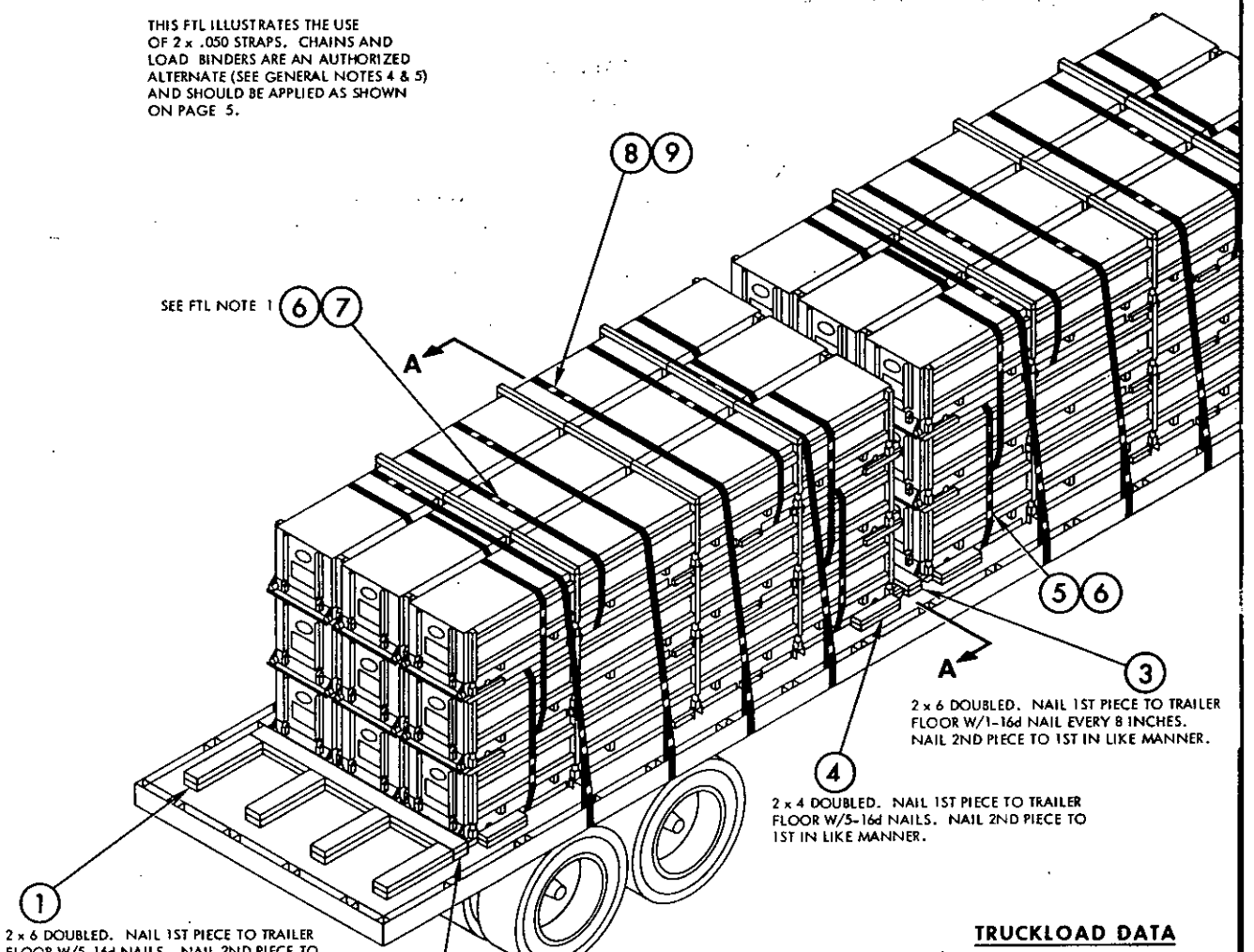
TABLE I

CONTAINER	DIMENSIONS	CUBE (CU.FT.)	TARE WEIGHT (LBS.)	GROSS WEIGHT (LBS.)	TRUCKLOAD WEIGHT (LBS.)
MK 372 MOD 4	202 1/2 L x 28 W x 28 7/16 H	93.4	679	2070	37,260
MK 372 MOD 6	202 1/2 L x 28 W x 28 7/16 H	93.4	700	2091	37,638
CNU-183/E	193 1/2 L x 28 W x 28 7/16 H	89.2	663	2033	36,594

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NOTE

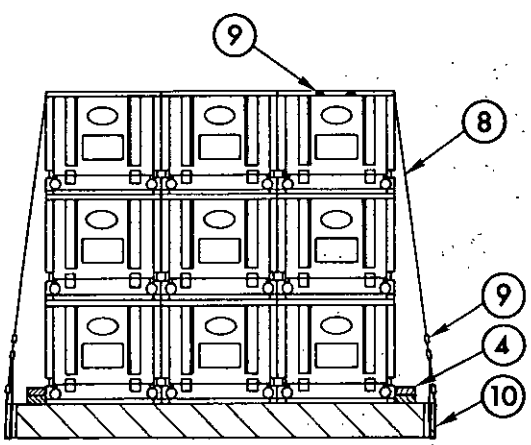
THIS FTL ILLUSTRATES THE USE OF 2 x .050 STRAPS. CHAINS AND LOAD BINDERS ARE AN AUTHORIZED ALTERNATE (SEE GENERAL NOTES 4 & 5) AND SHOULD BE APPLIED AS SHOWN ON PAGE 5.



TRUCKLOAD DATA

NUMBER OF UNITS. 18

WEIGHT. . . . . SEE TABLE I



SECTION A-A

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
10	STAKE POCKET PAD (STRAPPING)	2 x .050 x 24	16	SEE LOAD STRAPPING, PAGE 7		
9	SEAL	FOR 2 IN. STRAP	64	-	-	-
8	TIE DOWN STRAP	2 x .050 x 29 FT	8	SEE LOAD STRAPPING, PAGE 7		
7	CROSS STRAP	1 1/4 x .035 x 20 FT - 8	4	-	-	-
6	SEAL	FOR 1 1/4 STRAP	56	-	-	-
5	UNITIZING STRAP	1 1/4 x .035 x 16 FT	24	SEE PAGE 6		
4	SLEEPER	2 x 4 x 24	16	SEE FIELD NOTE		
3	CROSSMEMBER	2 x 6 x 90	2	SEE FIELD NOTE		
2	CROSSMEMBER	2 x 6 x 90	4	SEE FIELD NOTE		
1	END CLEAT	2 x 6 x 28	16	SEE FIELD NOTE		

LIST OF MATERIALS & NAILING DATA

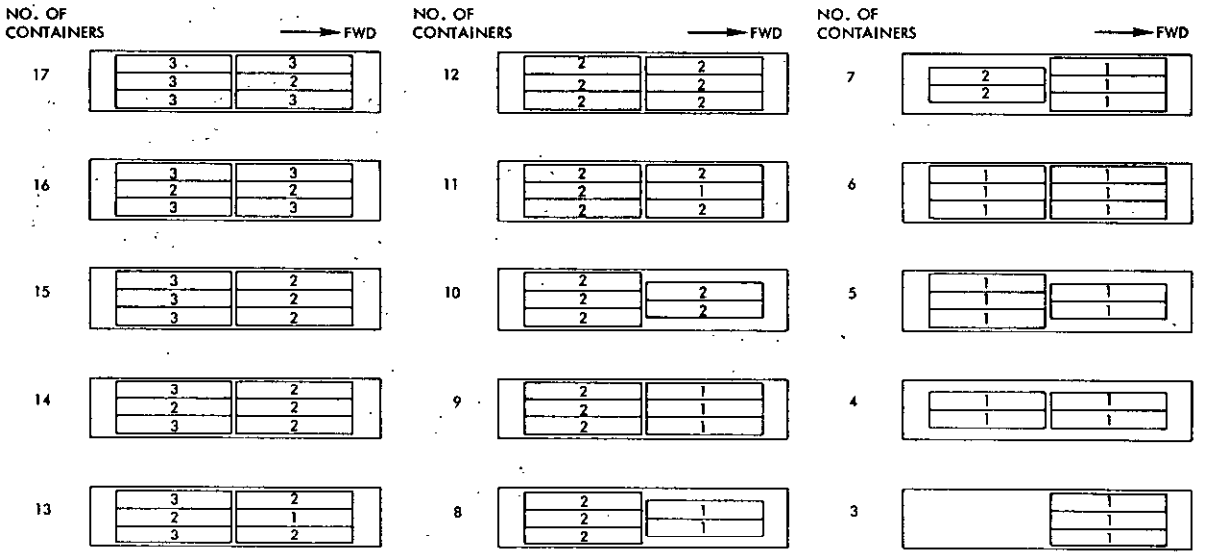
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LTL 40 FT. TRAILER (FLATBED)

1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF CONTAINERS TO BE SHIPPED FROM TABLE II AND POSITION CONTAINERS ON THE TRAILER ACCORDINGLY.

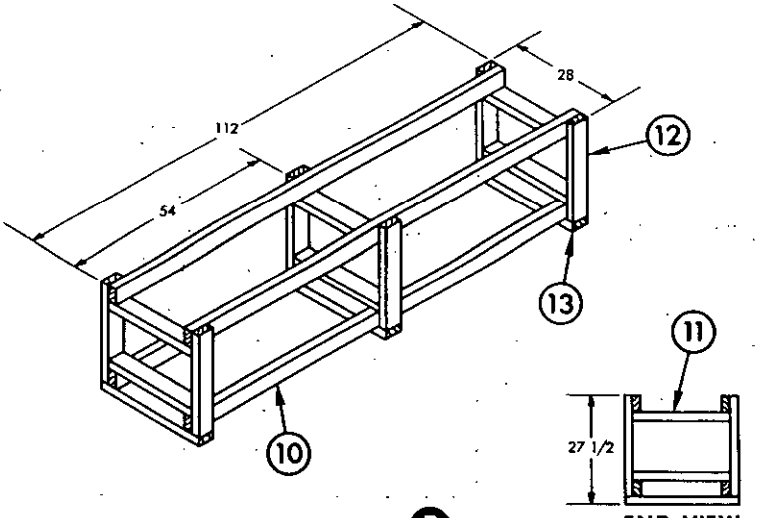
TABLE II

LOAD PATTERN *



*LOAD PATTERN SHOWS FLOOR PLAN FOR TRAILER. NUMBER INDICATES THE NUMBER OF CONTAINERS IN A LAYER.

2. WHEN THE CENTER OF THE TOP LAYER OF A STACK DOES NOT HAVE A CONTAINER (SEE LOAD PLAN FOR 14 CONTAINERS), THE FILLER ASSEMBLY DETAIL B SHALL BE USED. SECURE THE ASSEMBLY TO THE TIE DOWNS TO PREVENT LONGITUDINAL MOVEMENT.
3. ALL STACKS OF THREE HIGH CONTAINERS SHALL BE SECURED TO THE TRAILER WITH FOUR CHAINS (OR STRAPS). STACKS OF CONTAINERS TWO HIGH SHALL BE SECURED WITH THREE CHAINS (OR STRAPS).
4. THE LTL SHOWN ON PAGE 5 IS FOR FOURTEEN CONTAINERS AND SHOWS THE CORRECT BLOCKING, BRACING AND TIE DOWNS FOR A TRUCKLOAD OF THIS QUANTITY.
5. THE LTL PLAN SHOWN ON PAGE 5 DOES NOT LIMIT LTL SHIPMENTS TO THIS QUANTITY. ITS PRESENTATION IS INTENDED TO EXPLAIN THE CORRECT METHODS OF LOADING LTL SHIPMENTS.



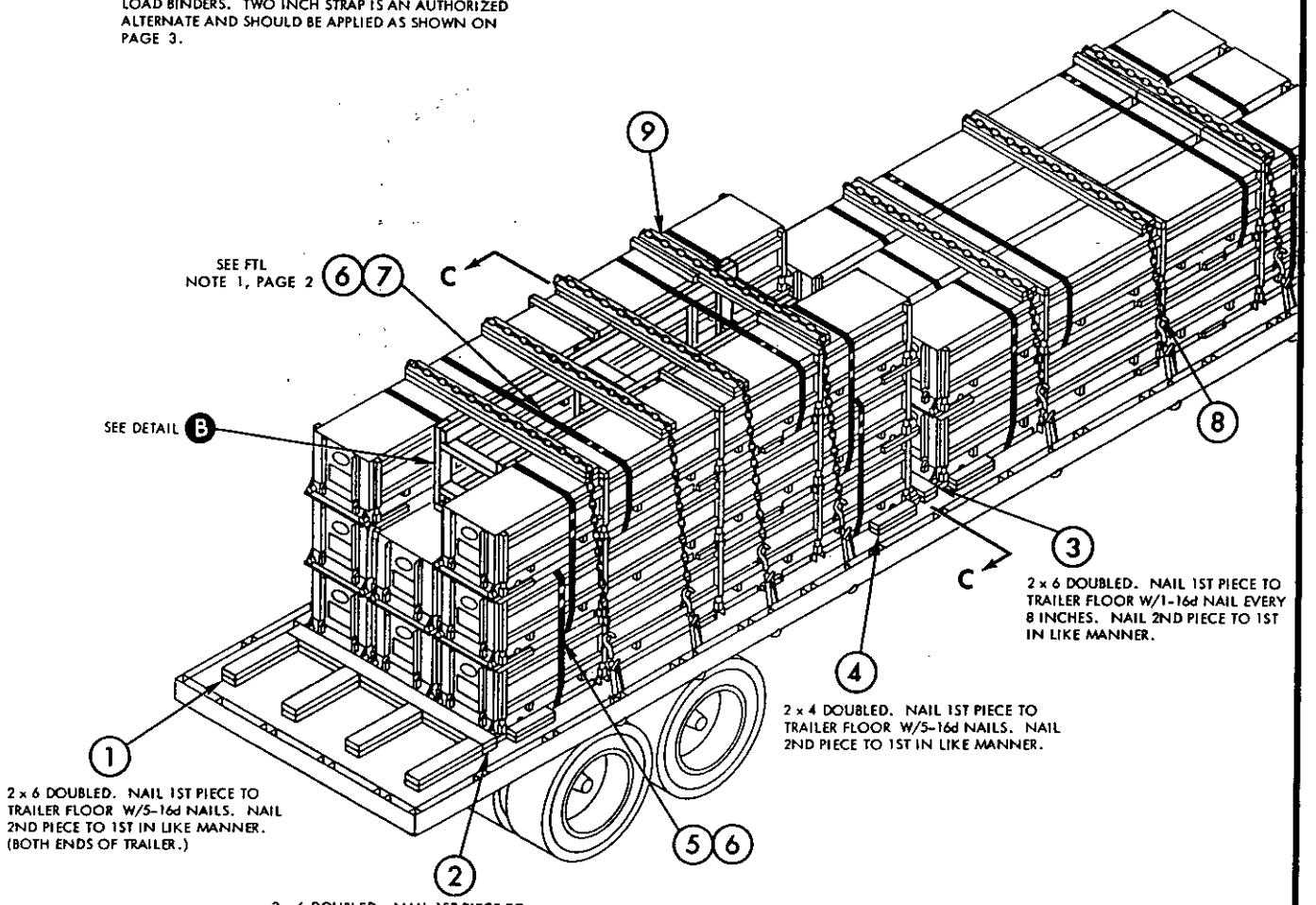
DETAIL B FILLER ASSEMBLY

END VIEW

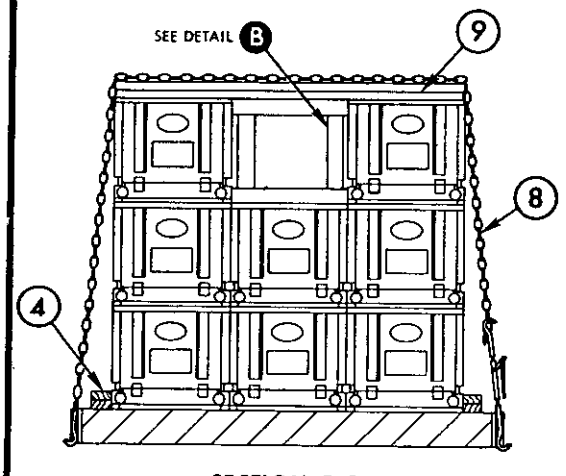
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NOTE

THIS LTL SHOWS THE PROPER APPLICATION OF CHAINS AND LOAD BINDERS. TWO INCH STRAP IS AN AUTHORIZED ALTERNATE AND SHOULD BE APPLIED AS SHOWN ON PAGE 3.



*MAY BE TWO PIECE.
JOIN WITH 2 x 4 x 24 SPLICE.



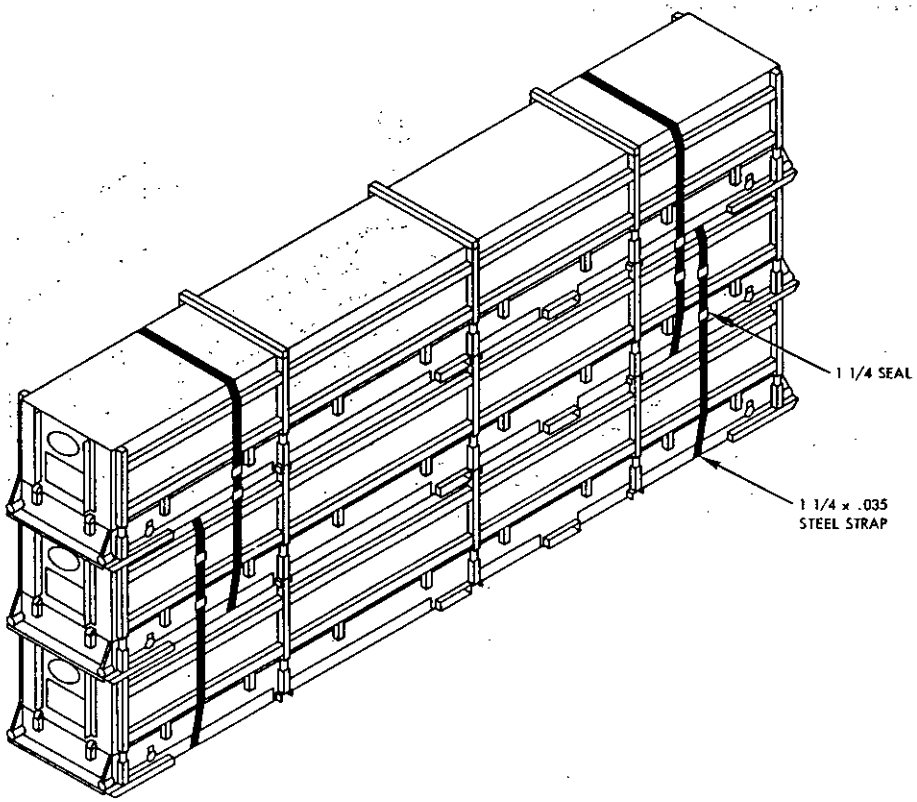
SECTION C-C

PIECE NO.	DESCRIPTION	SIZE	NO.PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
13	SUPPORT	2 x 4 x 28	3	10	2/JOINT	12d
12	VERTICAL	2 x 4 x 26	6	11	2/JOINT	16d
11	CROSS BRACE	2 x 4 x 25	6	10	2/JOINT	12d
10	HORIZONTAL PIECE	2 x 4 x 9 FT - 4*	4	12	2/JOINT	12d
9	PROTECTOR BOARD	2 x 6 x 82	14	DOUBLED & LAMINATED	2/FT	12d
8	CHAIN & LOAD BINDER	3/8	7	SEE GENERAL NOTES 4 & 5 PAGE 2		
7	CROSS STRAP	1 1/4 x .035 x 20 FT	4	-	-	-
6	SEAL	FOR 1 1/4 STRAP	40	-	-	-
5	UNITIZING STRAP	1 1/4 x .035 x 16 FT	16	SEE PAGE 6		
4	SLEEPER	2 x 4 x 24	16	SEE FIELD NOTE		
3	CROSSMEMBER	2 x 6 x 90	2	SEE FIELD NOTE		
2	CROSSMEMBER	2 x 6 x 90	4	SEE FIELD NOTE		
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LIST OF MATERIALS AND NAILING DATA

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UNITIZING AND HANDLING



CONTAINER UNITIZING AND BANDING PROCEDURE:

1. PLACE TWO OR MORE PIECES OF LUMBER CROSSWISE ON FLOOR OR GROUND NEXT TO MISSILE CONTAINER.
2. USING FORKLIFT TRUCK OR AN APPROPRIATE HOISTING DEVICE EQUIPPED WITH MK 77 MOD 3 HOISTING SLING, STACK THREE CONTAINERS ON THE LUMBER PIECES.
3. BE CERTAIN THAT NESTING FEATURES AND BEARING SURFACES ARE PROPERLY POSITIONED TO PREVENT SHIFTING IN LATERAL AND LONGITUDINAL DIRECTIONS.
4. SECURE BOTTOM CONTAINER TO CENTER CONTAINER WITH TWO STRAPS POSITIONED AS INDICATED ABOVE.
5. SEAL EACH STRAP WITH TWO SEALS AND DOUBLE CRIMP.
6. REPEAT THIS OPERATION TO SECURE CENTER CONTAINER TO TOP CONTAINER.

CONTAINER HANDLING PROCEDURE:

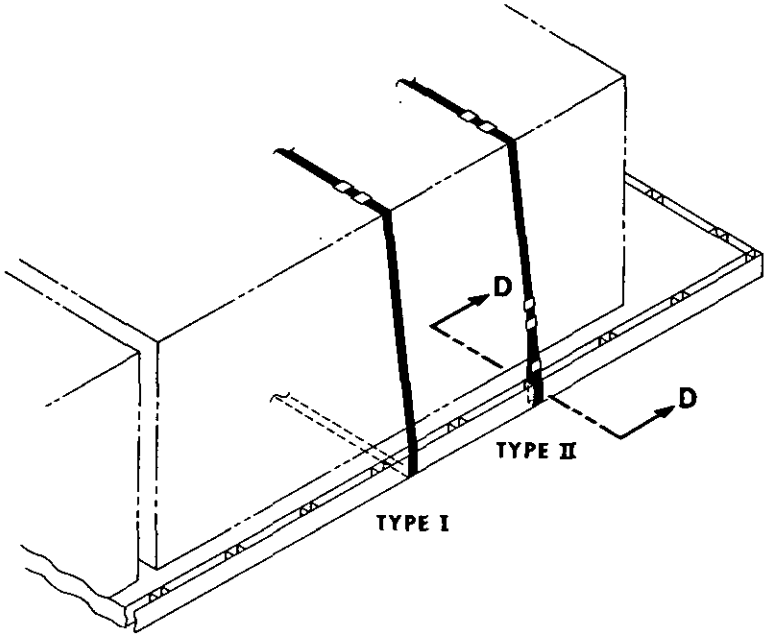
1. UNITIZED CONTAINERS MK 372 MOD 4 OR 6 OR CNU-183/E, FOR THE STANDARD ARM MISSILE ARE HANDLED AND POSITIONED ON FLATBED TRAILERS WITH A SUITABLE FORKLIFT TRUCK. IF NECESSARY CONTAINERS CAN BE LOADED ONE AT A TIME USING AN APPROPRIATE HOISTING DEVICE EQUIPPED WITH MK 77 MOD 3 HOISTING SLING, AND UNITIZED IN POSITION.

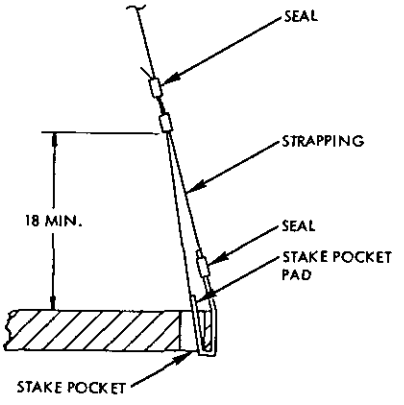
WARNING

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LOAD STRAPPING





SECTION D-D

THE FOLLOWING TWO METHODS OF TIE-DOWN STRAPPING ARE APPROVED FOR USE WITH THE SHIPMENT OF LOADS ON FLATBED TRAILERS. STEEL STRAPPING SHALL BE IN ACCORDANCE WITH FEDERAL SPECIFICATION QQ-S-781. STRAPPING SEALS SHALL HAVE A MINIMUM TENSILE STRENGTH EQUAL TO 75 PERCENT OF THE STRAP STRENGTH. FOR EACH OF THE TYPES ILLUSTRATED ABOVE IT IS PREFERRED TO POSITION, TENSION AND DOUBLE CRIMP THE STRAP SEALS AT THE TOP OF THE LOAD.

TYPE I - CONTINUOUS STRAPPING AROUND THE FLATBED AND THE LOAD.

TYPE II - THE STRAPPING IS SECURED TO THE STAKE POCKETS, ONE PIECE ON EACH SIDE OF THE TRAILER, AND IS BROUGHT UP OVER THE LOAD, TENSIONED, AND SEALED WITH THE DOUBLE-CRIMPED SEALS ON THE TOP. METHOD OF SECURING STRAPPING TO STAKE POCKET IS SHOWN IN SECTION D-D. THE SHORT END IS ON THE INSIDE AND IS SECURED WITH TWO DOUBLE-CRIMPED SEALS AT A MINIMUM OF 18 INCHES ABOVE THE TRAILER BED. A STAKE POCKET PAD (A SHORT PIECE OF THE SAME STRAPPING 18 INCHES LONG) IS INSERTED BETWEEN THE MAIN STRAP AND THE STAKE POCKET AND IS SECURED TO THE MAIN STRAP WITH A SEAL AS SHOWN.

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